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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITYANALYSIS

Approved By

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Prepared by the staffs of the International Monetary Fund and the International Development Association.[1]

Improved macroeconomics fundamentals, an increase in the discount rate and the migration to the 1993 System of National Accounts with an upward revision of the GDP series advocate in favor of a reclassification of the Democratic Republic of the Congo (DRC) debt sustainability rating from high risk to moderate risk of debt distress. However, the country is still vulnerable to shocks that could lead to a deterioration of some debt indicators. To maintain this rating the DRC will need to sustain fiscal discipline, create more fiscal space, including through a better mobilization of domestic revenues (both the natural resource and non-natural resource sectors), continue to borrow mainly on concessional terms, and improve its debt management capacity.

[1] This DSA was prepared by International Monetary Fund (IMF) staff with input from the World Bank, using the standard debt sustainability framework for low-income countries (LIC-DSA); see "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries."

BACKGROUND

1. The DRC's external debt decreased sharply since the country benefited from a substantial debt relief in 2010. The DRC reached the completion point under the Enhanced Heavily Indebted Poor Countries (HIPC) initiative and benefited from assistance under the Multilateral Debt Relief Initiative (MDRI) in July 2010. The ratio of public and publicly-guaranteed external debt (PPGE) to GDP was reduced from 75 percent at end 2009 to 22 percent in 2010 and represented 18 percent of GDP in 2013. Simultaneously the private external debt of the country increased from 2.1 percent of GDP in 2010 to 4.8 percent in 2013 but its exposure remained relatively limited.

2. The bulk of outstanding public external debt is owed to official creditors. In 2013, 60 percent of the debt was owed to multilateral creditors and only 4.5 percent to the bilateral official creditors. However the share of the debt owed to commercial creditors has marginally increased from 33.0 percent of the total public debt in 2010 to 35.5 percent at end-2013 (Text Table 1).

Text Table 1. Democratic Republic of the Congo: Total External Debt Stock, Central Government 2010–13

(millions of US$, unless otherwise indicated)

	2009	2010	2011	2012	2013	2013 (Percent)
	Act.				Est.	
Total	13705	4556	4629	4662	5196	100.0
Multilateral creditors	4986	2530	2655	2653	3121	60.1
Bilateral creditors	8130	247	242	242	233	4.5
Commercial creditors	589	1780	1733	1768	1842	35.5

Sources: Congolese authorities; and IMF staff estimates.

3. Terms of the most recently renegotiated mining investment for the Sino-Congolese (SCCA) joint venture (known as Sicomines) project do not change the external debt profile. In April 2008, the DRC signed a cooperation agreement with a consortium of Chinese enterprises involving a US$3.2 billion mining project and a set of public infrastructure projects totaling US$6 billion to be implemented in two phases. The Agreement was amended in October 2009 to exclude the second phase public infrastructure projects, leaving just a single phase totaling US$3 billion to be implemented over the period 2009–14. The amended agreement also limited the government's guarantee solely to the financing of the infrastructure projects. The 2012 debt sustainability analysis (DSA) was conducted based on the October 2009 agreement. Implementation of the public investment component stalled in 2012 as legislation envisaged under the Sicomines agreement was not acted upon by Parliament. In 2013, the Sicomines agreement was further amended and staff received in early 2014 information about the changes introduced. Based on the available information, the original financial terms of the agreement have been maintained and the public guarantee continues to be limited to the infrastructure projects totaling $3 billion. Recent passage of the legislation paved the way for the resumption of public investment program. The renegotiated agreement

envisages mining production to start in 2015, somewhat earlier than projected in 2009, but with a lower initial production capacity. However, over the project period, the total size of both the mining and the public investment projects remained unchanged.

4. One of the main change since the 2012 DSA is the revision of the GDP series following the migration to the 1993 System of National Accounts (SNA93). With technical assistance from the IMF, the national accounts were rebased to 2005 from 1987 together with a migration to the SNA93, resulting in an upward revision of nominal GDP of about 60 percent. The SNA better takes account of the informal economy and in this context the GDP reevaluation was widely attributed to the private sector, on the consumption and investment posts. Export and imports data have not been revised upward since estimates of parallel trade were already included in BOP data. As a result, the revision of the GDP series has a strong impact on the debt-to-GDP ratios, but not on the other debt ratios (Text Table 2).

Text Table 2. Democratic Republic of the Congo: Comparative Debt Ratios, 2012–17

	2012	2013	2014	2015	2016	2017
	Act.	Est.	Proj.			
PPGE debt to GDP ratio						
old DSA	32.3	34.7	36.5	36.8	37.4	36.5
new DSA	17.0	17.9	19.3	20.7	21.4	23.3
PPGE debt to exports ratio						
old DSA	56.5	56.5	56.9	55.3	58.5	51.5
new DSA	49.8	49.2	49.2	51.5	52.0	55.1
PPGE debt to revenue ratio						
old DSA	144.9	159.9	162.9	160.5	160.2	153.1
new DSA	113.8	137.9	138.3	147.4	145.6	156.8

Sources: Congolese authorities; and IMF staff estimates and projections.

5. As in the 2012 DSA, strong growth over the medium term is supported by large investments in mining and public infrastructure projects. Projections for mining production have been revised upward substantially over the short term based on updated information from major mining companies and the investment profile for the SCCA revised in line with the renegotiated agreement leading to a quicker accumulation of debt over the next few year. The strong growth in mining production should lead to stronger fiscal revenues in 2014 after lower than projected fiscal revenues in 2013. The export growth has been weaker than expected in the previous DSA for 2013 and is projected to increase smoothly over the medium term (Text Table 3 and Box 1).

6. Higher discount rate improves some indicators. Compared to the previous DSA the discount rate in the DSA used to calculate the present value (PV) of external debt has been adjusted upward from 4.0 percent to 5.0 percent, which decreases the present value of the debt.

Text Table 3. Democratic Republic of the Congo: Selected Indicators, 2012–17

	2012	2013	2014	2015	2016	2017
	Act.	Est.	Proj.			
Real GDP growth						
old DSA	7.2	8.2	6.4	7.3	5.1	8.0
new DSA	7.2	8.5	8.7	8.5	7.9	7.3
Revenues (excluding grants) growth						
old DSA	33.4	6.1	11.7	10.9	8.2	11.2
new DSA	38.6	-4.8	16.1	10.0	14.5	9.6
Overall fiscal deficit (percent of GDP)						
old DSA	-2.6	-3.4	-3.8	-3.5	-3.3	-2.8
new DSA	0.6	-1.7	-2.2	-1.7	-1.6	-3.0
Exports of goods and services growth						
old DSA	-5.6	17.2	12.8	12.6	2.0	20.7
new DSA	-8.5	16.1	4.7	5.8	9.7	6.4
Current account deficit (percent of GDP)						
old DSA	-12.5	-14.2	-15.4	-12.0	-15.9	-8.5
new DSA	-8.0	-10.1	-9.3	-8.8	-7.8	-7.4

Sources: Congolese authorities; and IMF staff estimates and projections.

Box 1. Democratic Republic of the Congo: Macroeconomic Assumptions for 2014–33

For the medium term (2014–18), projections are consistent with the macroeconomic framework of the 2014 Article IV Consultation, which assumes a strong growth driven by mining production and investments in infrastructures.

Long-term (2019–33) projections assume macroeconomic stability and a moderate growth on account of a slowdown in mining production partially offset by sustained increase in investments and a dynamic tertiary sector.

Real GDP growth is expected to increase from 8.5 percent in 2013 to 8.7 percent in 2014 and average about 7.5 percent during 2015–18. It should decline in the long run to 4.5 percent on average for the period 2019–33.

Average Inflation (measured by the GDP deflator in dollars) is 0.6 percent in 2013 and is expected to increase to 2.1 percent on average over the medium-long run.

The overall fiscal balance displayed a deficit of 1.7 percent of GDP in 2013 and is expected to slightly increase in the medium-long run to 3.1 percent on average due to a reduction of grants as percent of GDP.

The non-interest current account deficit is projected to decrease from 10.0 percent of GDP in 2013 to 6.6 percent in 2018. In the long term it should keep narrowing to 1.5 percent of GDP by 2033, due to the slowdown of FDIs (which have high import content) and the development of the agricultural sector.

Financing: The authorities are expected to rely increasingly on private financial flows (especially FDIs) to finance the current account deficit and the public investments. FDIs are expected to decrease in percent of GDP from 7.4 percent in 2013 to 3.2 percent on average in the medium-long run, but should remain the main financing source of the current account deficit. Aid flows represented 5.6 percent of GDP in 2013 out of which 1.2 percent were concessional loans. Over the medium term, this ratio would decline marginally to 5.3 percent by 2017 and further to 1.3 percent by 2033.

EXTERNAL DEBT SUSTAINABILITY RESULTS

7. The most extreme shock leads to a significant breach of the debt to exports ratio and a slighter breach of the debt to GDP and to revenue ratios, while the other indicators are expected to remain below their thresholds. With the shock in exports the PV of debt-to-export ratio would increase significantly from 34 percent in 2013 to 178 percent in 2015 and remain well above the threshold of 100 percent for the main part of the projection period (Figure 1 and Tables 1 and 2).

8. The debt ratios are consistently lower in the historical scenario than the baseline. This is because the non-interest current account deficit has been lower on average from 2005 to 2012 than projected for the period 2014–33 in the baseline scenario, and the net inflows of FDI higher. Thus applying the historical numbers leads to lower gross financing needs and a lower present value of external public debt. And since the net inflow of FDIs has been higher than the non-interest current account deficit from 2003 to 2012, at an average of 6.9 percent of GDP compared to 2.9 percent, the PV of the external debt declines to low single digit in the historic scenario during the projected period. Indeed, assuming sustained larger FDIs than the current account deficit—as is the assumption under the historical scenario—gross financing needs decline and nominal value of foreign public debt becomes negligible.

9. Overall, compared with the previous DSA, the current results suggest that the DRC's risk of debt distress has decreased from high to moderate. This is largely on account of improved macroeconomic fundamentals, the significant upward revision of the GDP series and the increase of the discount rate to 5 percent.

PUBLIC DSA RESULTS

10. The DRC's (estimated) domestic debt is primarily domestic arrears. These include arrears owed to domestic creditors, "social debts" (e.g., wages owed to past government employees), arrears owed to state-owned enterprises (SOEs), arrears owed to suppliers and contractors, and unpaid judgments and awards stemming from judicial decisions. Not all of the arrears identified in the last audit of domestic arrears have been certified and many claims on the government may be at least partly offset by liabilities owed to the government. Based on the last audit conducted, the debt management unit estimates that approximately US$1400 million (4.5 percent of 2013 GDP) would be valid after further assessment. Out of this amount authorities plan to obtain a discount from creditors of at least 50 percent, leaving an amount of US$700 million to clear; this is the estimate used in this DSA. The baseline includes the expected clearance of all arrears by 2024. Consequently, while the PV of total public debt-to-GDP ratio is 2.6 percent of GDP higher in 2013 than that of the PV of external debt-to-GDP ratio in the baseline scenario, the difference declines over time and disappears after 2019 (Figure 2).

11. Although the fixed primary balance scenario outcome is quite similar to the baseline, the historical scenario leads to a substantial fall in debt levels similar to historic scenario under the external debt analysis. The debt ratio in the historical scenario is lower than the baseline partly as a result of the higher primary fiscal balance (average of 1.2 percent of GDP in 2005–12) used in the historical scenario projections when compared to the average projected primary deficit in the baseline scenario

(average -3.1 percent of GDP over 2014–33, Table 3 and 4). Thus, if on average, real GDP grows as fast as it did over the last decade and the government pursues a fiscal policy that generates smaller primary deficits than those forecast under the baseline, it could generate sufficient savings such that the country's net debt (debt minus assets) becomes negative after 2023.

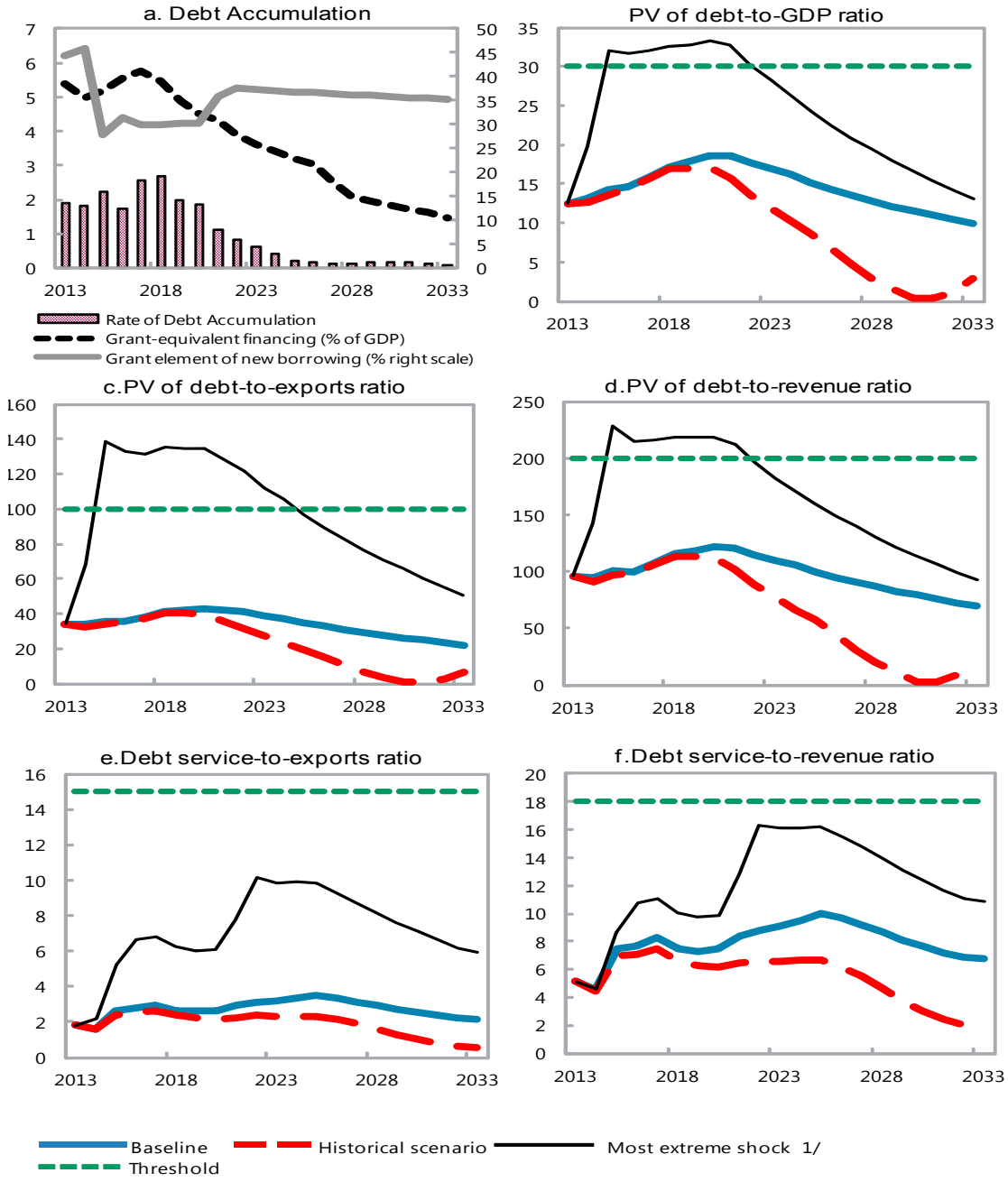
12. Public debt remains sustainable even under the shock scenarios. The most extreme shock is a shock on GDP growth.[2] This implies a raise of the PV of the debt-to-GDP ratio from 14.7 percent to a maximum of 23.0 percent in 2021 that remains below the benchmark of the public debt ratio of 38 percent. The impact is also significant on the PV of the debt-to-exports ratio, which rises to 131.8 percent of GDP over the long run (64 percentage points of GDP higher than under the baseline).

CONCLUSION

13. On account of the better macroeconomic fundamentals, an upward revision of the GDP series, and a higher discount rate the DRC's debt sustainability rating is upgraded from a high risk to a moderate risk of debt distress. However, a significant portion of the debt stock and repayment profile in the DRC is attributable to the publicly-guaranteed Sicomines infrastructure loans and the Sicomines mining investments which should be carefully monitored. Under the current framework, not only is the country still vulnerable to a drop in commodity prices that would reduce growth and drive the PV of debt ratios above their thresholds, but a sizable enough price shock could undermine the economics of the mining project that is supposed to pay off the infrastructure loans. This would shift the repayment burden for much or the entire loan outstanding on the government. It is urgent that the DRC improves its compilation of debt data, including Sicomines data, and strengthens its debt risk analysis capabilities. Meanwhile, the DRC should continue to take a cautious approach to external borrowing.

[2] The real GDP is assumed to grow at its historical average minus one standard deviation in 2014–15.

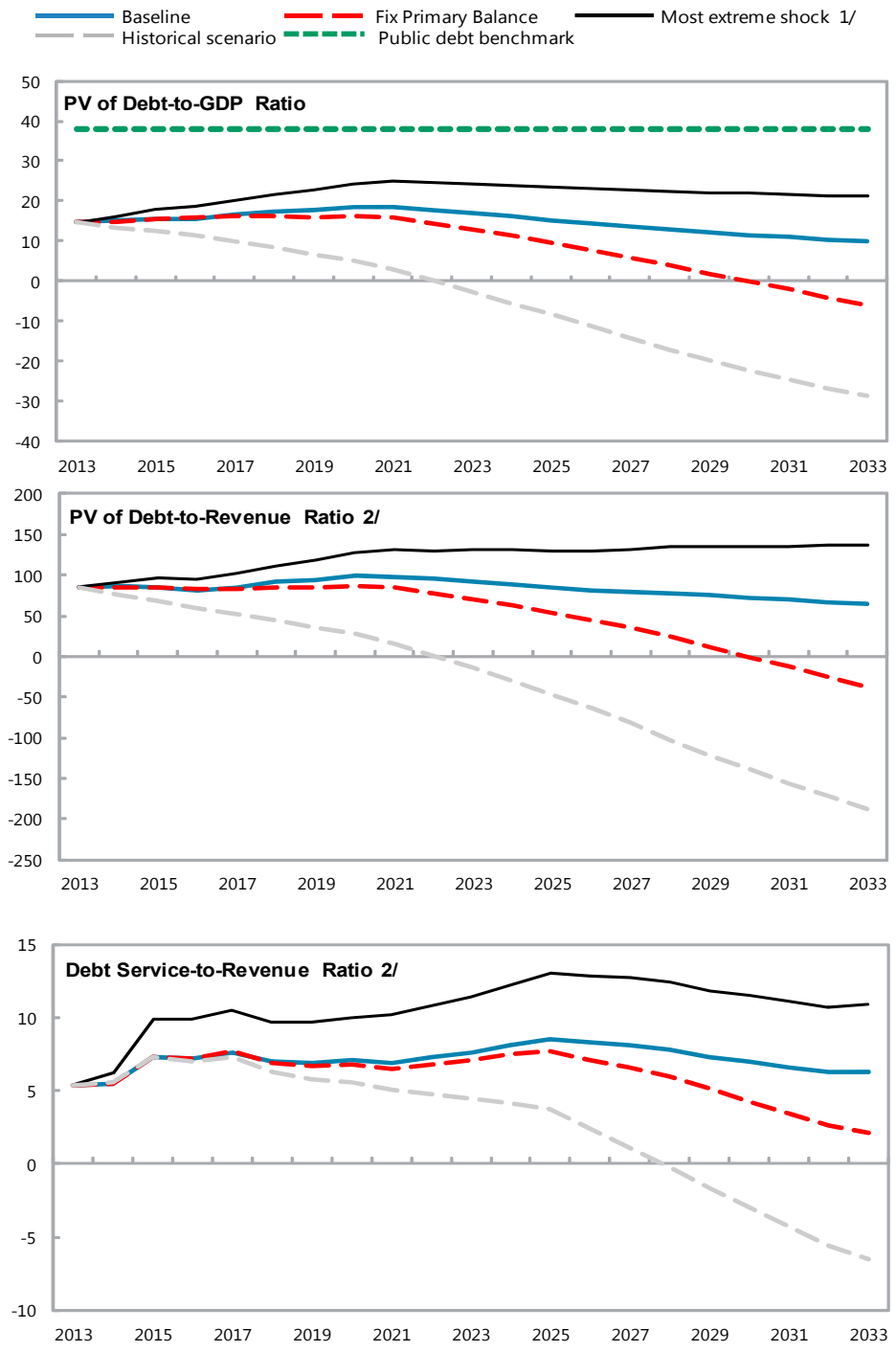
Figure 1. Democratic Republic of the Congo: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013–33[1]



Sources: Congolese authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figures b, c, d, e and f, it corresponds to a Exports shock.

Figure 2. Democratic Republic of the Congo: Indicators of Public Debt under Alternative Scenarios, 2013–33[1]



Sources: Congolese authorities; and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.
2/ Revenues are defined inclusive of grants.

Table 1. Democratic Republic of the Congo: External Debt Sustainability Framework, Baseline Scenario, 2010–33[1]

(in percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**24.5**	**21.3**	**20.0**			**23.0**	**26.0**	**29.2**	**30.4**	**32.1**	**33.0**		**24.1**	**14.2**	
of which: public and publicly guaranteed (PPG)	22.4	19.2	16.9			18.0	19.4	21.0	21.8	23.6	25.2		24.1	14.2	
Change in external debt	-59.2	-3.2	-1.3			3.0	3.1	3.1	1.2	1.6	0.9		-1.5	-0.8	
Identified net debt-creating flows	-12.2	-4.5	-2.6			1.1	0.5	-0.1	-0.8	-0.3	0.0		1.9	-1.3	
Non-interest current account deficit	**4.7**	**5.8**	**7.9**	**2.9**	**4.2**	**10.0**	**9.2**	**8.5**	**7.4**	**6.9**	**6.6**		**5.9**	**1.5**	5.4
Deficit in balance of goods and services	5.5	4.8	5.9			6.1	2.0	-0.3	-1.1	-1.9	-1.0		-1.8	-9.5	
Exports	43.6	45.5	34.1			36.4	39.1	40.2	41.3	42.3	41.8		43.9	45.4	
Imports	49.1	50.3	40.0			42.5	41.1	39.9	40.1	40.4	40.8		42.1	35.9	
Net current transfers (negative = inflow)	-3.9	-3.3	-2.9	-3.9	1.0	-5.3	-4.7	-4.1	-4.2	-4.1	-4.0		-3.2	-2.0	-2.9
of which: official	-3.9	-3.3	-2.9			-5.3	-4.7	-4.1	-4.2	-4.1	-4.0		-3.2	-2.0	
Other current account flows (negative = net inflow)	3.2	4.3	4.9			9.1	11.9	12.9	12.7	12.9	11.6		10.8	13.1	
Net FDI (negative = inflow)	**-7.3**	**-7.0**	**-7.9**	**-6.9**	**1.8**	**-7.4**	**-6.9**	**-6.9**	**-6.5**	**-5.6**	**-5.2**		**-3.2**	**-2.6**	-3.2
Endogenous debt dynamics 2/	**-9.6**	**-3.4**	**-2.7**			**-1.5**	**-1.7**	**-1.6**	**-1.7**	**-1.6**	**-1.4**		**-0.8**	**-0.3**	
Contribution from nominal interest rate	0.2	0.1	0.1			0.1	0.1	0.4	0.4	0.4	0.5		0.4	0.2	
Contribution from real GDP growth	-5.2	-1.4	-1.3			-1.6	-1.9	-2.0	-2.1	-2.0	-1.9		-1.2	-0.5	
Contribution from price and exchange rate changes	-4.5	-2.0	-1.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-47.0**	**1.3**	**1.4**			**1.8**	**2.6**	**3.2**	**2.1**	**2.0**	**0.9**		**-3.4**	**0.5**	
of which: exceptional financing	-22.2	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	14.7			17.4	19.8	22.4	23.3	24.3	24.9		17.0	9.9	
In percent of exports	...	...	43.0			47.7	50.6	55.8	56.5	57.6	59.6		38.6	21.7	
PV of PPG external debt	**...**	**...**	**11.6**			**12.5**	**13.2**	**14.2**	**14.7**	**15.9**	**17.2**		**17.0**	**9.9**	
In percent of exports	**...**	**...**	**33.9**			**34.2**	**33.8**	**35.4**	**35.6**	**37.6**	**41.0**		**38.6**	**21.7**	
In percent of government revenues	**...**	**...**	**77.5**			**95.8**	**94.9**	**101.3**	**99.6**	**107.1**	**115.1**		**109.6**	**69.3**	
Debt service-to-exports ratio (in percent)	**1.3**	**1.5**	**1.9**			**1.8**	**1.6**	**2.6**	**2.8**	**2.9**	**2.7**		**3.2**	**2.1**	
PPG debt service-to-exports ratio (in percent)	**1.3**	**1.5**	**1.9**			**1.8**	**1.6**	**2.6**	**2.8**	**2.9**	**2.7**		**3.2**	**2.1**	
PPG debt service-to-revenue ratio (in percent)	**4.8**	**5.4**	**4.4**			**5.1**	**4.6**	**7.5**	**7.7**	**8.3**	**7.4**		**9.1**	**6.8**	
Total gross financing need (Millions of U.S. dollars)	-411	-112	198			981	944	914	773	1058	1135		2,664	-46	
Non-interest current account deficit that stabilizes debt ratio	63.9	9.0	9.2			7.1	6.1	5.3	6.2	5.3	5.7		7.3	2.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.1	6.9	7.2	6.0	1.3	8.5	8.7	8.5	7.9	7.3	6.4	7.9	5.1	3.6	4.5
GDP deflator in US dollar terms (change in percent)	5.7	9.0	7.4	6.0	6.3	0.6	-0.4	0.6	1.1	1.4	2.1	0.9	2.5	2.8	2.5
Effective interest rate (percent) 5/	0.2	0.5	0.6	2.4	1.5	0.6	0.5	1.6	1.6	1.6	1.6	1.3	1.9	1.6	1.7
Growth of exports of G&S (US dollar terms, in percent)	79.9	21.5	-13.7	29.0	42.3	16.6	16.5	12.1	11.9	11.5	7.5	12.7	9.9	8.4	7.7
Growth of imports of G&S (US dollar terms, in percent)	48.6	19.4	-8.5	26.7	32.2	16.1	4.7	5.8	9.7	9.6	9.6	9.3	7.5	2.1	6.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	44.4	45.8	27.9	31.3	30.0	29.9	34.9	37.3	35.2	35.5
Government revenues (excluding grants, in percent of GDP)	12.1	12.4	14.9			13.0	13.9	14.0	14.7	14.8	14.9		15.5	14.2	15.0
Aid flows (in Billions of US dollars) 7/	1,948	1,486	1,921			1,668	1,759	1,836	2,120	2,232	2,283		2,259	1,724	
of which: Grants	1,848	1,336	1,421			1,318	1,137	1,486	1,770	1,882	1,933		1,920	1,480	
of which: Concessional loans	100	150	500			350	622	350	350	350	350		339	244	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.4	5.0	5.2	5.6	5.8	5.5		3.6	1.5	2.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			80.9	73.9	66.7	71.9	65.7	65.1		75.8	72.2	73.5
Memorandum items:															
Nominal GDP (Millions of US dollars)	20,500	23,870	27,483			30,012	32,516	35,477	38,690	42,117	45,737		65,427	127,777	
Nominal dollar GDP growth	13.2	16.4	15.1			9.2	8.3	9.1	9.1	8.9	8.6	8.9	7.7	6.5	7.1
PV of PPG external debt (Millions of US dollars)			3,187			3,712	4,255	4,979	5,592	6,583	7,714		10,917	12,404	
(PVt-PVt-1)/GDPt-1 (in percent)						1.9	1.8	2.2	1.7	2.6	2.7	2.2	0.6	0.1	0.5
Gross workers' remittances (Millions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	11.6			12.5	13.2	14.2	14.7	15.9	17.2		17.0	9.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	33.9			34.2	33.8	35.4	35.6	37.6	41.0		38.6	21.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.9			1.8	1.6	2.6	2.8	2.9	2.7		3.2	2.1	

Sources: Congolese authorities; and IMF staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Democratic Republic of the Congo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33

(in percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
				PV of debt-to GDP ratio				
Baseline	12	13	14	15	16	17	**17**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	12	13	14	15	16	17	**12**	3
A2. New public sector loans on less favorable terms in 2013-2033 2/	12	14	15	17	19	21	**23**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	12	14	15	16	17	18	**18**	10
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	12	20	32	32	32	33	**28**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	12	13	14	15	16	17	**17**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	12	16	18	19	20	21	**19**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	12	17	22	22	23	24	**22**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	12	19	20	21	22	24	**24**	14
				PV of debt-to-exports ratio				
Baseline	34	34	35	36	38	41	**39**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	34	32	34	35	37	40	**27**	6
A2. New public sector loans on less favorable terms in 2013-2033 2/	34	36	38	40	44	49	**52**	37
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	34	33	35	35	37	40	**38**	21
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	34	68	138	133	132	135	**112**	50
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	34	33	35	35	37	40	**38**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	34	40	46	45	47	50	**44**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	34	46	60	59	59	63	**55**	27
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	34	33	35	35	37	40	**38**	21
				PV of debt-to-revenue ratio				
Baseline	96	95	101	100	107	115	**110**	69
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	96	90	97	98	106	113	**77**	21
A2. New public sector loans on less favorable terms in 2013-2033 2/	96	100	109	112	125	139	**148**	119
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	96	98	107	105	113	122	**116**	73
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	96	142	228	214	216	219	**182**	92
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	96	94	101	99	106	114	**109**	68
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	96	112	131	127	132	139	**126**	74
B5. Combination of B1-B4 using one-half standard deviation shocks	96	119	154	147	152	158	**139**	78
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	96	134	143	140	151	162	**154**	97

Table 2. Democratic Republic of the Congo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33 (concluded)

(in percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
	Debt service-to-exports ratio							
Baseline	2	2	3	3	3	3	**3**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	2	3	3	2	**2**	1
A2. New public sector loans on less favorable terms in 2013-2033 2/	2	2	3	3	3	3	**4**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	2	3	3	3	3	**3**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	2	5	7	7	6	**10**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	2	3	3	3	3	**3**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	3	3	3	3	**4**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	3	4	4	3	**5**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	2	3	3	3	3	**3**	2
	Debt service-to-revenue ratio							
Baseline	5	5	7	8	8	7	**9**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	5	4	7	7	7	7	**7**	2
A2. New public sector loans on less favorable terms in 2013-2033 2/	5	5	7	8	9	8	**11**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	5	5	8	8	9	8	**10**	7
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	5	5	9	11	11	10	**16**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	5	5	8	8	8	8	**9**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	5	5	8	8	9	8	**11**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	8	9	10	9	**12**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	5	7	11	11	12	11	**13**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	33	**33**	33

Sources: Congolese authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Democratic Republic of the Congo: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–33

(in percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections									
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	26.9	22.8	19.8			20.4	21.3	22.3	22.7	24.2	25.5		24.1	14.2	
of which: foreign-currency denominated	22.4	19.2	16.9			18.0	19.4	21.0	21.8	23.6	25.2		24.1	14.2	
Change in public sector debt	-62.7	-4.1	-3.1			0.6	0.9	1.1	0.4	1.5	1.3		-1.0	-0.8	
Identified debt-creating flows	-26.2	-4.5	-4.9			-1.4	-0.8	-1.4	-1.5	0.0	0.6		0.7	3.6	
Primary deficit	-4.4	1.1	-0.7	-1.2	1.4	1.6	2.0	1.3	1.2	2.6	3.1	2.0	2.9	4.4	3.4
Revenue and grants	21.1	18.0	20.1			17.4	17.4	18.2	19.3	19.3	19.1		18.4	15.4	
of which: grants	9.0	5.6	5.2			4.4	3.5	4.2	4.6	4.5	4.2		2.9	1.2	
Primary (noninterest) expenditure	16.8	19.0	19.4			19.0	19.5	19.5	20.5	21.9	22.2		21.3	19.8	
Automatic debt dynamics	-17.3	-4.1	-2.8			-1.4	-1.4	-1.4	-1.4	-1.4	-1.5		-1.3	-0.7	
Contribution from interest rate/growth differential	-10.8	-3.8	-2.9			-1.6	-1.5	-1.4	-1.5	-1.4	-1.4		-1.3	-0.7	
of which: contribution from average real interest rate	-4.9	-2.0	-1.4			0.0	0.1	0.2	0.2	0.1	0.0		-0.1	-0.2	
of which: contribution from real GDP growth	-5.9	-1.7	-1.5			-1.5	-1.6	-1.7	-1.6	-1.5	-1.5		-1.2	-0.5	
Contribution from real exchange rate depreciation	-6.4	-0.3	0.1			0.2	0.1	0.1	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	-4.6	-1.5	-1.4			-1.6	-1.4	-1.4	-1.3	-1.2	-1.0		-0.9	-0.1	
Privatization receipts (negative)	-4.6	-1.5	-1.4			-1.6	-1.4	-1.4	-1.3	-1.2	-1.0		-0.9	-0.1	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-36.5	0.4	1.8			2.0	1.7	2.5	1.9	1.6	0.7		-1.6	-4.4	
Other Sustainability Indicators															
PV of public sector debt	...	...	14.4			14.8	15.0	15.6	15.6	16.5	17.5		17.0	9.9	
of which: foreign-currency denominated	...	...	11.6			12.5	13.2	14.2	14.7	15.9	17.2		17.0	9.9	
of which: external	...	...	11.6			12.5	13.2	14.2	14.7	15.9	17.2		17.0	9.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-3.6	2.0	0.2			2.6	3.0	2.6	2.6	4.1	4.4		4.3	5.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	71.8			85.1	86.3	85.5	80.9	85.5	91.3		92.1	64.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	96.7			113.9	108.0	111.0	106.0	111.2	117.2		109.6	69.3	
of which: external 3/	...	...	77.5			95.8	94.9	101.3	99.6	107.1	115.1		109.6	69.3	
Debt service-to-revenue and grants ratio (in percent) 4/	3.8	5.2	4.5			5.4	5.4	7.3	7.2	7.6	6.9		7.6	6.3	
Debt service-to-revenue ratio (in percent) 4/	6.6	7.5	6.0			7.2	6.8	9.5	9.5	9.9	8.9		9.1	6.8	
Primary deficit that stabilizes the debt-to-GDP ratio	58.3	5.2	2.4			1.0	1.2	0.2	0.8	1.1	1.8		3.8	5.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.1	6.9	7.2	6.0	1.3	8.5	8.7	8.5	7.9	7.3	6.4	7.9	5.1	3.6	4.5
Average nominal interest rate on forex debt (in percent)	0.2	0.5	0.6	2.5	1.5	0.7	0.7	2.2	2.2	2.2	2.2	1.7	1.9	1.6	1.8
Average real interest rate on domestic debt (in percent)	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Real exchange rate depreciation (in percent, + indicates depreciation	-8.7	-1.7	0.4	-0.1	14.3	1.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	17.4	10.3	7.5	16.9	9.1	0.7	1.3	2.9	4.0	4.8	5.6	3.2	6.0	6.4	6.0
Growth of real primary spending (deflated by GDP deflator, in percer	39.2	21.3	9.1	7.4	13.0	6.4	11.3	8.7	13.4	14.7	7.9	10.4	4.5	2.9	3.7
Grant element of new external borrowing (in percent)	...	...	...	...	...	44.4	45.8	27.9	31.3	30.0	29.9	34.9	37.3	35.2	...

Sources: Country authorities; and IMF staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Democratic Republic of the Congo: Sensitivity Analysis for Key Indicators of Public Debt, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
	PV of Debt-to-GDP Ratio							
Baseline	14.8	15	16	16	17	17	17	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	15	13	13	11	10	9	-3	-29
A2. Primary balance is unchanged from 2013	15	15	16	16	16	16	13	-6
A3. Permanently lower GDP growth 1/	15	15	16	16	17	18	19	15
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	15	16	18	19	20	22	24	21
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	15	14	14	14	15	16	16	9
B3. Combination of B1-B2 using one half standard deviation shocks	15	14	14	14	16	18	20	18
B4. One-time 30 percent real depreciation in 2014	15	20	19	19	19	20	19	13
B5. 10 percent of GDP increase in other debt-creating flows in 2014	15	22	22	22	22	23	22	13
	PV of Debt-to-Revenue Ratio 2/							
Baseline	85	86	85	81	85	91	92	64
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	85	76	68	58	52	44	-14	-188
A2. Primary balance is unchanged from 2013	85	85	85	82	84	85	70	-38
A3. Permanently lower GDP growth 1/	85	87	86	82	87	94	101	97
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	85	91	96	94	102	112	131	137
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	85	79	75	72	77	83	85	59
B3. Combination of B1-B2 using one half standard deviation shocks	85	79	74	74	82	92	109	113
B4. One-time 30 percent real depreciation in 2014	85	113	107	98	99	103	105	85
B5. 10 percent of GDP increase in other debt-creating flows in 2014	85	124	120	112	115	120	117	83
	Debt Service-to-Revenue Ratio 2/							
Baseline	5	5	7	7	8	7	8	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	6	7	7	7	6	4	-7
A2. Primary balance is unchanged from 2013	5	5	7	7	8	7	7	2
A3. Permanently lower GDP growth 1/	5	5	7	7	8	7	8	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	5	6	8	8	8	8	9	11
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	5	5	7	7	7	7	7	6
B3. Combination of B1-B2 using one half standard deviation shocks	5	6	7	7	8	7	8	9
B4. One-time 30 percent real depreciation in 2014	5	6	10	10	11	10	11	11
B5. 10 percent of GDP increase in other debt-creating flows in 2014	5	5	8	8	8	8	10	8

Sources: Congolese authorities; and IMF staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.